

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that the DNS service that the website is attempting to reach is unavailable to service the request to access the website.

This is based on the results of the network analysis, which show that the ICMP protocol reply returned the error message stating that "udp port 53 unreachable". Issues with performing the DNS protocol further proves because the plus sign after the query identification number 35084 indicates flags with the UDP message and the "A?" symbol indicates flags with performing DNS protocol operations.

Since port 53 is associated with DNS traffic we know that the issue is with the DNS server.

The most likely issue is that the DNS server is not responding which can be concluded from the error message about port 53.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred today at 1:24PM.

Explain how the IT team became aware of the incident: Customers notified the IT team that they received an error message stating "destination port unreachable" when they attempted to visit the website yummyrecipesforme.com.

Explain the actions taken by the IT department to investigate the incident: Packet sniffing tests were conducted using tcpdump to investigate the issue.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): According to the log file, we found that DNS port 53 was unreachable

Note a likely cause of the incident: DNS server could be down or the traffic to port 53 is blocked by the firewall. Therefore the DNS server could be down due to a successful DOS attack from malicious individuals or a misconfiguration preventing traffic requests on port 53.

